LA SALLE AGKT 2627

Redox Reactions & Electrolysis

化學筆記

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I. Redox Reactions — Basic Concepts 5 items

A redox reaction is a reaction in which oxidation and reduction take place at the same time. One species loses electrons while another gains them.

Three ways to identify oxidation and reduction

- Oxygen / hydrogen transfer: gain of oxygen or loss of hydrogen = oxidation; loss of oxygen or gain of hydrogen = reduction
- Electron transfer: loss of electrons = oxidation; gain of electrons = reduction (OIL RIG — Oxidation Is Loss, Reduction Is Gain)
- Oxidation number: an increase in O.N. = oxidation; a decrease in O.N. = reduction

Example — $Na(s) + H^+(aq) \rightarrow Na^+(aq) + \frac{1}{2}H_2(g)$

- Na loses an electron → oxidized (O.N. 0 → +1); Na is the reducing agent
- H^+ gains an electron → reduced (O.N. +1 → 0); H^+ is the oxidizing agent

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II. Oxidation Numbers (O.N.) 6 items

The oxidation number (O.N.) of an element shows how many electrons its atoms have lost or gained compared with the free element. An increase in O.N. means oxidation; a decrease means reduction.

Rules for assigning oxidation numbers

- Atoms in a free element have O.N. = 0 (e.g. O_2, Fe, Cl_2)
- A monatomic ion has O.N. equal to its charge (e.g. Na^+ = +1, O^{2-} = -2)
- Fluorine is always -1
- Oxygen is -2, except in peroxides (-1, e.g. H_2O_2) and OF_2 (+2)
- Hydrogen is +1, except in metal hydrides (-1, e.g. NaH)
- The sum of oxidation numbers is 0 for a neutral compound, or equal to the charge for an ion

Worked example — O.N. of Mn in MnO_4^- and MnO_2

Let the O.N. of Mn be x, and that of O be -2. For MnO_4^-: x + 4(-2) = -1, so x = +7. For MnO_2: x + 2(-2) = 0, so x = +4.

Species	Element	O.N.
MnO_4^-	Mn	+7
MnO_2	Mn	+4
$Cr_2O_7^{2-}$	Cr	+6
CO_3^{2-}	C	+4
NO_3^-	N	+5
SO_4^{2-}	S	+6

III. Writing & Balancing Redox Equations 9 items

Half-equations

A half-equation shows only the oxidation part or the reduction part of a redox reaction, with electrons on the product side (oxidation) or the reactant side (reduction).

$$SO_2(g) + 2H_2O(l) \rightarrow SO_3^{2-}(aq) + 4H^+(aq) + 2e^- \text{ (S: +4 → +6, loses 2 } e^-\text{)}$$

$$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l) \text{ (Mn: +7 → +2, gains 5 } e^-\text{)}$$

Steps for balancing with half-equations

- Write the two half-equations — balance atoms first (add H_2O, then H^+), then balance charges (add e^-)
- Multiply the half-equations by suitable factors so that the numbers of electrons are equal
- Add the two equations together and cancel the electrons on both sides
- Check that atoms and charges balance

Example — Fe^{2+} with acidified $KMnO_4$

$$MnO_4^-(aq) + 8H^+(aq) + 5Fe^{2+}(aq) \rightarrow Mn^{2+}(aq) + 4H_2O(l) + 5Fe^{3+}(aq)$$

Check with oxidation numbers: the total increase in O.N. must equal the total decrease in O.N.

IV. Common Oxidizing & Reducing Agents (DSE) 8 items

Common oxidizing agents

- Acidified $KMnO_4$: $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ (purple → colourless)

- Alkaline $KMnO_4$: $MnO_4^- + 2H_2O + 3e^- \rightarrow MnO_2 + 4OH^-$ (purple → brown)
- Acidified $K_2Cr_2O_7$: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$ (orange → green)
- Chlorine: $Cl_2 + 2e^- \rightarrow 2Cl^-$ (pale green → colourless)
- Dilute/concentrated HNO_3 and concentrated H_2SO_4

Common reducing agents

- Metals: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$; $Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$
- Sulphur dioxide: $SO_2 + 2H_2O \rightarrow SO_4^{2-} + 4H^+ + 2e^-$ (decolourizes iodine solution)
- Iodide: $2I^- \rightarrow I_2 + 2e^-$ (colourless → brown)
- H_2S, CO and H_2

Redox titration

- $MnO_4^- + 5Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$ — no indicator is needed because MnO_4^- is self-indicating (purple colour just disappears at the end point)

Concentrated acids as oxidizing agents — the sugar experiment

- When concentrated H_2SO_4 is dropped onto white sugar, a black spongy mass of carbon forms and a pungent gas (SO_2) is evolved — the acid dehydrates the sugar and then oxidizes it
- When concentrated HNO_3 is dropped onto white sugar, a pungent brown gas (NO_2) is evolved — HNO_3 acts as the oxidizing agent

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V. The Full Redox Reaction List — HNO_3, H_2SO_4 & Cl_2 43 items

Nitric acid — concentration decides its behaviour

- Very dilute (~0.2 M): a typical dilute acid — only acidic properties
- Dilute (~2 M): an oxidizing acid — strong oxidizing agent + acidic properties
- Concentrated (~16 M): a strong oxidizing agent; decomposes in sunlight ($4HNO_3 \rightarrow 4NO_2 + O_2 + 2H_2O$) — stored in brown bottles

Dilute HNO_3 + metals (except Pt and Au)

Metal(s) + dilute $HNO_3(aq) \rightarrow$ metal nitrate(aq) + NO(g) + $H_2O(l)$

$3Cu(s) + 8HNO_3(aq) \rightarrow 3Cu(NO_3)_3(aq) + 2NO(g) + 4H_2O(l)$

Ionic: $3Cu(s) + 8H^+(aq) + 2NO_3^-(aq) \rightarrow 3Cu^{2+}(aq) + 2NO(g) + 4H_2O(l)$

- The colourless NO reacts with O_2 at the tube mouth: $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$
- Observation: Cu dissolves to give a blue solution; a brown pungent gas appears at the mouth

Hot concentrated HNO_3 + metals (except Pt and Au)

$$Metal(s) + conc.\ HNO_3(aq) \rightarrow metal\ nitrate(aq) + NO_2(g) + H_2O(l)$$

$$Cu(s) + 4HNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2NO_2(g) + 2H_2O(l)$$

$$\text{Ionic: } Cu(s) + 4H^+(aq) + 2NO_3^-(aq) \rightarrow Cu^{2+}(aq) + 2NO_2(g) + 2H_2O(l)$$

- Brown pungent NO_2 is formed directly (no need for air)

Hot concentrated HNO_3 + non-metals

$$C(s) + 4HNO_3(aq) \rightarrow CO_2(g) + 4NO_2(g) + 2H_2O(l)$$

$$S(s) + 6HNO_3(aq) \rightarrow H_2SO_4(aq) + 6NO_2(g) + 2H_2O(l)$$

- The SO_2 first formed is further oxidized by concentrated HNO_3 to sulphate + NO_2

Concentrated HNO_3 + reducing agents

$$SO_2(g) + 2H^+(aq) + 2NO_3^-(aq) \rightarrow SO_4^{2-}(aq) + 2NO_2(g) + H_2O(l)$$

Concentrated sulphuric acid (~18 M) — oxidizing AND dehydrating

- Dilute (2 M) H_2SO_4: a typical acid — only acidic properties
- Concentrated: oxidizing acid + dehydrating agent (corrosive due to both)

Hot concentrated H_2SO_4 + metals (except Pt and Au)

$$Metal(s) + conc.\ H_2SO_4(l) \rightarrow metal\ sulphate(aq) + SO_2(g) + H_2O(l)$$

$$Cu(s) + 2H_2SO_4(l) \rightarrow CuSO_4(aq) + SO_2(g) + 2H_2O(l)$$

$$\text{Ionic: } Cu(s) + 2H^+(aq) + H_2SO_4(l) \rightarrow Cu^{2+}(aq) + SO_2(g) + 2H_2O(l)$$

Hot concentrated H_2SO_4 + non-metals

$$C(s) + 2H_2SO_4(l) \rightarrow CO_2(g) + 2SO_2(g) + 2H_2O(l)$$

$$S(s) + 2H_2SO_4(l) \rightarrow 3SO_2(g) + 2H_2O(l)$$

Concentrated H_2SO_4 as a dehydrating agent (see also Chemical Tests & Drying Agents)

$CuSO_4 \cdot 5H_2O(s) \rightarrow CuSO_4(s) + 5H_2O(l)$ (blue → white)

$C_{12}H_{22}O_{11}(s) \rightarrow 12C(s) + 11H_2O(l)$ (sugar charring — ⬛ black carbon)

Chlorine — preparation

$OCl^- (aq) + Cl^- (aq) + 2H^+ (aq) \rightarrow Cl_2 (g) + H_2O(l)$ (dilute acid + chlorine bleach)

$MnO_2 (s) + 4HCl (aq) \rightarrow MnCl_2 (aq) + Cl_2 (g) + 2H_2O(l)$ (MnO_2 + conc. HCl)

- Dry with concentrated H_2SO_4; collect by downward delivery or a gas syringe

Halogen displacement — oxidizing power $F_2 > Cl_2 > Br_2 > I_2$

$Cl_2 (g/aq) + 2Br^- (aq) \rightarrow 2Cl^- (aq) + Br_2 (aq)$ (⬜ colourless → 🟧 orange)

$Cl_2 (g/aq) + 2I^- (aq) \rightarrow 2Cl^- (aq) + I_2 (aq)$ (⬜ colourless → 🟫 brown)

$Br_2 (aq) + 2I^- (aq) \rightarrow 2Br^- (aq) + I_2 (aq)$ (Br_2 cannot oxidize Cl^-)

Chlorine + sodium hydroxide — disproportionation

$Cl_2 (g) + 2NaOH (aq) \rightarrow NaCl (aq) + NaOCl (aq) + H_2O(l)$ (cold dilute → bleach)

$Cl_2 (g) + 2OH^- (aq) \rightarrow Cl^- (aq) + OCl^- (aq) + H_2O(l)$

$3Cl_2 (g) + 6NaOH (aq) \rightarrow 5NaCl (aq) + NaClO_3 (aq) + 3H_2O(l)$ (hot concentrated → chlorate)

- Disproportionation: Cl is oxidized and reduced at the same time in these reactions

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VI. Electrolysis — Basic Principles 6 items

Electrolysis is the decomposition of an electrolyte by an electric current. It takes place in an electrolytic cell.

The electrolytic cell

- An electrolyte conducts electricity by the movement of mobile ions — it must be molten or aqueous

- Cathode (−): connected to the negative terminal of the power supply; reduction occurs (cations gain electrons)
- Anode (+): connected to the positive terminal; oxidation occurs (anions lose electrons)

Conductors vs electrolytes

- Molten $PbBr_2$ is a typical electrolyte — it shows ionic property, conducting by mobile ions
- Cl_2 (g), Br_2 (g) and I_2 (g) are neutral molecular gases — there are no mobile ions, so they do not conduct

> $PbBr_2$ (l) → Pb(l) + Br_2 (g) (cathode: $Pb^{2+} + 2e^- \rightarrow Pb$; anode: $2Br^- \rightarrow Br_2 + 2e^-$)

VII. Electrolysis of Molten Ionic Compounds 5 items

A molten ionic compound contains only TWO kinds of ions, so there is no competition — every cation moves to the cathode and every anion moves to the anode.

Example — electrolysis of molten $PbBr_2$

- Cathode (−): silvery, shiny beads of molten lead form ($Pb^{2+} + 2e^- \rightarrow Pb$)
- Anode (+): reddish-brown vapour of bromine is given off ($2Br^- \rightarrow Br_2 + 2e^-$)

Other examples

- Molten NaCl: sodium metal at the cathode; chlorine gas at the anode
- Molten Al_2O_3 (in molten cryolite): aluminium at the cathode; oxygen at the anode

VIII. Electrolysis of Aqueous Solutions 4 items

Selective discharge — which ions are discharged?

- Position in the reactivity series: the LOWER the position of the metal, the more readily its cation is discharged ($Cu^{2+} > H^+ > Na^+, K^+$)
- Concentration: the more concentrated an anion is, the more readily it is discharged (concentrated NaCl gives Cl_2 at the anode instead of O_2)
- Nature of the electrode: with copper electrodes in $CuSO_4$ (aq), the copper anode itself dissolves ($Cu \rightarrow Cu^{2+} + 2e^-$)

Standard examples (inert electrodes)

- Dilute NaCl (aq): H_2 at the cathode; O_2 at the anode; the remaining solution becomes alkaline (NaOH)
- Concentrated NaCl (aq): H_2 at the cathode; pale green Cl_2 at the anode (bleaches moist litmus); NaOH remains

- $CuSO_4$ (aq) with carbon electrodes: reddish-brown Cu deposited at the cathode; O_2 at the anode; the blue colour fades
- $CuSO_4$ (aq) with copper electrodes: the anode dissolves and Cu deposits at the cathode — the blue colour stays unchanged (basis of electrorefining)

IX. Electroplating 5 items

Electroplating uses electrolysis to cover an object with a thin layer of metal, for protection and decoration.

Set-up rules

- The object to be plated is the CATHODE
- The plating metal is the ANODE
- The electrolyte contains ions of the plating metal

Example — silver-plating a spoon

- Cathode: the spoon — $Ag^+ + e^- \rightarrow Ag$ deposits a shiny silver layer
- Anode: a piece of silver — $Ag \rightarrow Ag^+ + e^-$ (the anode dissolves, keeping the Ag^+ concentration constant)
- Electrolyte: $AgNO_3$ (aq)

X. Colours of Common Substances & Reactions 10 items

Colours of ions in aqueous solutions

Ion / solution	Colour
Cu^{2+}	Blue
Fe^{2+}	Pale green
Fe^{3+}	Yellow-brown
$Cr_2O_7^{2-}$	Orange
CrO_4^{2-}	Yellow
MnO_4^-	Purple
Mn^{2+}	Very pale pink
Ni^{2+}	Green
Co^{2+}	Pink

Colours of common precipitates

Precipitate	Colour
$Cu(OH)_2$	Blue
$Fe(OH)_2$	Green (turns brown in air — oxidized to $Fe(OH)_3$)
$Fe(OH)_3$	Rust-brown
$Al(OH)_3$, $Zn(OH)_2$, $Pb(OH)_2$	White
$AgCl$	White (darkens in sunlight)
$AgBr$	Pale yellow
AgI	Yellow
$BaSO_4$, $CaCO_3$	White
PbI_2	Bright yellow

Colours and properties of common gases

- Cl_2 — pale green / yellowish green; pungent, toxic; bleaches moist litmus
- NO_2 — brown; pungent, toxic
- Br_2 vapour — reddish-brown
- I_2 vapour — purple/violet
- NO — colourless, but turns brown on mixing with air ($2NO + O_2 \rightarrow 2NO_2$)
- SO_2 — colourless; pungent, choking smell
- NH_3 — colourless; pungent; turns damp red litmus blue
- H_2, O_2, N_2, CO_2 — colourless and odourless (H_2 burns with a pop sound)

Colours of common solids

- Cu — reddish-brown solid; CuO — black
- $CuSO_4 \cdot 5H_2O$ — blue crystals; anhydrous $CuSO_4$ — white powder
- Rust $Fe_2O_3 \cdot xH_2O$ — reddish-brown; $FeSO_4 \cdot 7H_2O$ — pale green crystals
- S (sulphur) — yellow solid; MnO_2 and C (carbon) — black solids
- $K_2Cr_2O_7$ crystals — orange; $KMnO_4$ crystals — purple

Colour changes in redox reactions — quick reference

Species change	Colour change
MnO_4^- → Mn^{2+} (acidified)	Purple → very pale pink (appears colourless when dilute)
MnO_4^- → MnO_2 (alkaline)	Purple → brown solid
$Cr_2O_7^{2-}$ → Cr^{3+}	Orange → green
Br_2 → Br^-	Orange → colourless
I_2 → I^-	Brown → colourless
Fe^{2+} → Fe^{3+}	Pale green → yellow-brown
Cu^{2+} → Cu (displacement)	Blue solution → reddish-brown deposit
NO → NO_2 (in air)	Colourless → brown
$CuSO_4 \cdot 5H_2O$ → $CuSO_4$ (heated)	Blue → white

XI. Chemical Tests & Drying Agents 16 items

Tests for gases

Gas	Test	Positive result
H_2	Lighted splint	Burns with a squeaky pop
O_2	Glowing splint	The splint relights
CO_2	Bubble into limewater, $Ca(OH)_2$ (aq)	Limewater turns milky ($CaCO_3$); dissolves in excess CO_2
Cl_2	Damp litmus paper	Turns red then bleaches white; pale green gas
NH_3	Damp red litmus paper	Turns blue; white fumes with concentrated HCl
HCl (g)	Ammonia rod (dipped in concentrated NH_3 (aq))	White fumes (NH_4Cl)
SO_2	Acidified $K_2Cr_2O_7$ (aq) / acidified $KMnO_4$ (aq)	Orange → green; purple → colourless
H_2S	Lead(II) ethanoate paper	Turns black (PbS)
NO_2	Observe the gas	Brown, pungent
Water vapour	Anhydrous $CuSO_4$ / cobalt(II) chloride paper	White → blue; blue → pink

Flame tests for cations

Cation	Flame colour
Li^+	Crimson
Na^+	Golden yellow
K^+	Lilac (view through cobalt blue glass)
Ca^{2+}	Brick red
Ba^{2+}	Apple green
Cu^{2+}	Blue-green

Tests for cations (precipitation)

Cation	NaOH (aq), a little → in excess	NH_3 (aq), a little → in excess
Cu^{2+}	Blue precipitate, insoluble in excess	Blue precipitate, dissolves in excess to a deep blue solution
Fe^{2+}	Green precipitate, turning brown on standing	Green precipitate
Fe^{3+}	Rust-brown precipitate	Rust-brown precipitate
Al^{3+}	White precipitate, dissolves in excess (colourless)	White precipitate, insoluble in excess
Pb^{2+}	White precipitate, dissolves in excess (colourless)	White precipitate, insoluble in excess
Zn^{2+}	White precipitate, dissolves in excess (colourless)	White precipitate, dissolves in excess (colourless)
Ca^{2+}	Faint white precipitate, insoluble in excess	No precipitate
Ag^{+}	Brown precipitate (Ag_2O)	— (test with HCl instead: white AgCl, insoluble in HNO_3)
NH_4^{+}	No precipitate; on warming, pungent NH_3 is evolved (turns damp red litmus blue)	—

Tests for anions

Anion	Test	Positive result
Cl^-	$AgNO_3$ (aq) + dilute HNO_3	White precipitate (AgCl), dissolves in dilute NH_3 (aq)
Br^-	$AgNO_3$ (aq) + dilute HNO_3	Cream precipitate (AgBr), dissolves only in concentrated NH_3 (aq)
I^-	$AgNO_3$ (aq) + dilute HNO_3	Yellow precipitate (AgI), insoluble in NH_3 (aq)
CO_3^{2-}	Add dilute HCl; pass the gas into limewater	Effervescence; limewater turns milky (dissolves in excess CO_2)
SO_4^{2-}	$BaCl_2$ (aq) or $Ba(NO_3)_2$ (aq) + dilute acid	White precipitate ($BaSO_4$)
SO_3^{2-}	Add dilute HCl; pass the gas into acidified $K_2Cr_2O_7$ (aq)	SO_2 evolved; orange → green
NO_3^- (1)	Aluminium foil (or Devarda's alloy) + NaOH (aq), warm	Pungent NH_3 evolved; turns damp red litmus blue
NO_3^- (2)	$FeSO_4$ (aq), then carefully layer concentrated H_2SO_4	A brown ring forms at the interface

Common acid-base indicators

Indicator	Acid	Neutral	Alkali
Litmus	Red	Purple	Blue
Phenolphthalein	Colourless (pH < 8.3)	Colourless	Deep pink (pH > 10)
Methyl orange	Red (pH < 3.1)	Orange (pH 3.1–4.4)	Yellow (pH > 4.4)

Halogen displacement — observations in an organic solvent

Halogen added	With Br^- (aq)	With I^- (aq)
Cl_2	Organic layer turns orange-red (Br_2)	Organic layer turns purple/violet (I_2)
Br_2	— (no reaction)	Organic layer turns purple/violet (I_2)

Oxidizing power: $F_2 > Cl_2 > Br_2 > I_2$; reducing power of halide ions: $I^- > Br^- > Cl^- > F^-$. Use 1,1,1-trichloroethane (or heptane) as the organic solvent — the halogen dissolves in it and shows its colour clearly.

Drying agents for gases

Drying agent	Gases to be dried
Concentrated H_2SO_4 (acidic)	Neutral gases and acidic gases (H_2, O_2, N_2, CO_2, SO_2, HCl, Cl_2)
Anhydrous $CaCl_2$ (s)	All gases except NH_3
CaO (s) (basic)	Neutral and alkaline gases — especially NH_3

- Drying = removal of water that is NOT chemically combined; dehydrating = removal of chemically combined water — a drying agent need not be a dehydrating agent (concentrated H_2SO_4 is the only dehydrating agent learnt)
- Anhydrous Na_2SO_4 and $MgSO_4$ are used to dry organic liquids (e.g. organic solvents)
- NH_3 CANNOT be dried by concentrated H_2SO_4 or $CaCl_2$ — they react: $2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$